Mahmoud Alyosify

AI & Machine Learning Engineer

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SUMMARY

AI and machine learning engineer with an **MSc in Artificial Intelligence from Queen's University, Canada (GPA 4.30 / 4.30)**. Builds LLM, RAG and machine learning systems and takes them from the first experiment to a tested, working service. Currently an **AI Engineer Intern at Injaz Tech**, building a RAG assistant over company documents. Master's thesis: a training-free method that makes open LLMs generate **47.6% fewer tokens** with no answer cut off. Earlier, supported the backend of Egypt's national toll network and taught AI and computer science to **10,000+ learners**.

SKILLS

AI & LLMs: LLMs, RAG, LangChain, LangGraph, AI agents, fine-tuning (LoRA, QLoRA), prompt engineering, Hugging Face Transformers

Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, computer vision, NLP, reinforcement learning (PPO, DQN)

Data: Python, SQL, Pandas, NumPy, PySpark, FAISS, MongoDB, feature engineering

Deployment: FastAPI, Docker, AWS, ONNX, GitHub Actions, pytest, Git, Linux

Evaluation: LLM-as-judge, MT-Bench, AlpacaEval, retrieval metrics (top-1, MRR), ablation studies, statistical testing

Programming: Python, SQL, C++, C, C#, .NET, JavaScript, Bash

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group) — Maadi, Cairo

- Built a RAG assistant that answers staff questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20 through measured design changes.
- Covered the service with **136 automated tests** (pytest) and an evaluation check that compares each new version with the last.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.
- Teach the **AI for Business** module of the same program, on a separate engagement.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

MinimalLM — Master's thesis 2026
Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Answer quality held: LLM-judge score 8.04/10 against 8.42 for the unchanged models (cutting the originals to the same length scores 6.46); the saving holds in six other languages (45.6%).

HORUS Sentinel — multi-agent intelligence platform 2026
Python, LLM agents, knowledge graph, self-hosted LLM

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.

Horus-OSINT — fine-tuned LLM on AWS 2026
Llama-3-8B, QLoRA, Hugging Face, AWS

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

SimCLR-Vision-SSL — self-supervised image model 2026
PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

Also: RL-Cloud-Autoscaler (PPO and DQN) · SCRAP (satellite collision risk, 48+ hours ahead) · AudioShield (deepfake audio detection) · Automated Multimodal Agent (PDF to narrated slides).

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI)
- AWS Academy: Machine Learning Foundations, ML for NLP, Data Engineering
- Microsoft Certified: Data Analyst Associate

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)